

Image Restoration by Inverse Problem with Kernel Estimation

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Plan

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- Motion blur occurs when camera or scene motion affects the exposure.
- Variational and Bayesian approaches historically dominate blind deblurring.
- Deep learning achieves strong results but requires large datasets and heavy computation.
- We focus on an interpretable, training-free variational pipeline based on spectral priors.

- Estimate the blur kernel from the blurred image using anisotropic power spectrum modelling.
- Recover the sharp image through TV-regularised deconvolution solved by Split-Bregman.
- Pipeline:
 - spectral modulus estimation of the kernel,
 - phase reconstruction under support and positivity constraints,
 - TV-based restoration with tuned hyperparameters.

On considère :

$$v = u * k + b$$

- u : sharp latent image.
- k : non-negative, normalized blur kernel with compact support.
- b : additive impulse noise.
- Ill-posed inverse problem requiring prior assumptions.
- We assume spatially invariant linear motion blur.

$$|\widehat{u}(\xi n_\theta)|^2 \approx \frac{1}{\varepsilon + c_\theta \|\xi\|^2}$$

$$\Rightarrow |\widehat{D_\theta u}(\xi n_\theta)|^2 = |\xi|^2 |\widehat{u}(\xi n_\theta)|^2 = \frac{|\xi|^2}{\varepsilon + c_\theta \|\xi\|^2}$$

- Natural images follow an anisotropic power-law decay.
- Taking directional derivatives amplifies high frequencies.
- This allows isolating the blur kernel in the spectrum of $D_\theta v$.

Whitening and Directional Derivatives

$$D_{\theta}v = (D_{\theta}u) * k + D_{\theta}b$$

$$\widehat{D_{\theta}v}(\xi n_{\theta}) = i(\xi n_{\theta} \cdot n_{\theta}) \widehat{u}(\xi n_{\theta}) \widehat{k}(\xi n_{\theta}) + \widehat{D_{\theta}b}(\xi n_{\theta})$$

- Whitening: D_{θ} flattens the spectrum of natural images.
- The remaining directional variations come from the blur kernel.
- Enables isolating $|\widehat{k}|^2$ in the autocorrelation domain.

$$\begin{aligned}\mathbb{E} \left(|\widehat{D_\theta v}(\xi n_\theta)|^2 \right) &= |\xi|^2 |\widehat{u}(\xi n_\theta)|^2 |\widehat{k}(\xi n_\theta)|^2 + \mathbb{E} \left(|\widehat{D_\theta b}|^2 \right) \\ &= c_\theta |\widehat{k}(\xi n_\theta)|^2 + \mu_\theta\end{aligned}$$

- After whitening, the residual spectrum becomes linear in $|\widehat{k}|^2$.
- μ_θ is estimated using the kernel normalization constraint:

$$\int_{\mathbb{R}} P_\theta(h)(x) dx = 1.$$

- Allows reconstruction of $|\widehat{k}(\xi)|^2$ for all θ .

Autocorrelation Deconvolution with compensation filter

$$R_{D_\theta v} = R_{D_\theta u} * R_k + R_{D_\theta b}$$

- Autocorrelation of $D_\theta v$ gives access to the width of the blur in direction θ .
- A compensation filter deconvolves camera noise and intrinsic blur using CG iterations.
- Negative lobes near the center invalidate the compensation and revert to the raw autocorrelation.

- Initial support s_θ estimated from minima of compensated autocorrelation.
- Support is smoothed using the Lipschitz constraint:

$$|s_{\theta_i} - s_{\theta_j}| \leq \kappa |\theta_i - \theta_j|.$$

- Ensures coherent support across all projection directions.

$$F(R(D_\theta v))(\xi) = c_\theta |\hat{k}(\xi n_\theta)|^2 + \mu_\theta$$
$$\Rightarrow |\hat{k}(\xi n_\theta)|^2 = \frac{F(R(D_\theta v))(\xi) - \mu_\theta}{c_\theta}$$

- Full modulus $|\hat{k}(\xi)|$ recovered by interpolating directional slices.
- Noise bias removed using low-frequency estimates of μ_θ .
- Produces a stable modulus usable for phase retrieval.

- Random phases $\phi(\xi)$ sampled to reconstruct candidate kernels:

$$k = \mathcal{F}^{-1} \left(|\widehat{k}(\xi)| e^{i\phi(\xi)} \right)$$

- Constraints applied: positivity, compact support, normalization.
- Best candidate chosen by TV-deconvolution PSNR score.

$$\hat{u} = \arg \min_u \frac{1}{2} \|k * u - v\|_2^2 + \lambda \|\nabla u\|_1$$

Split-Bregman iterations:

$$u^{k+1} = \arg \min_u \frac{1}{2} \|k * u - v\|_2^2 + \frac{\gamma}{2} \|\nabla u - d^k + b^k\|_2^2$$

$$d^{k+1} = \text{shrink}(\nabla u^{k+1} + b^k, \lambda/\gamma)$$

$$b^{k+1} = b^k + (\nabla u^{k+1} - d^{k+1})$$

- Solved in Fourier domain using symmetric or circular boundary conditions.
- TV prior preserves edges and limits ringing.
- Convergence when $\|u^{k+1} - u^k\| < \eta$.

Estimation Principle

- Analysis of the directional spectrum of the image.
- Use of directional derivatives.
- Recovery of the Fourier transform modulus of the kernel.
- Kernel reconstruction via random phases.

Types of Experiments Conducted

- Validation with known vs estimated blur.
- Robustness to noise.
- Influence of boundary conditions.
- Effect of kernel size.
- Tests on real images.